

REMOTE-CONTROLLED 6-CAMERA CCTV SWITCHER

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Closed-circuit television (CCTV) uses video cameras to transmit signals to specific monitors. It differs from broadcast television in that the signal is not openly transmitted, though it may employ point-to-point wireless links. CCTV is often used for surveillance in areas that need security,

auto (the switcher selects the cameras one by one and holds videos of each camera till the preprogrammed time)

5. Auto mode: You can change 'on' time of each camera (hold time) individually through the remote. Time can be programmed from 0 to 255 seconds.

6. Camera bypass: You can set any camera in bypass mode by entering 0-second 'on' time for cameras which you want to bypass or don't want to watch during auto mode.

7. Audio: One-way audio. In this model, you can hear the audio from the camera selected by the switcher.

8. Video: Two video-out RCA sockets with 75-ohm impedance 1Vp-p are available at the rear panel of the master unit. You can connect two monitors in these sockets.

9. Other features: Wide input voltage range, long-range infrared remote commander, microcontroller-based design supported with CMOS digital

integrated circuits, liquid-crystal display to monitor the status and bilateral digital MOS switches for switching the analogue video/audio signals from cameras

At the heart of the circuit is Microchip's PIC16F72 microcontroller, which controls all the functions described in the design. Its pin configuration is shown in Fig. 1.

The circuit

Fig. 2 shows the main control circuit of the camera switcher. In this circuit, PIC16F72 microcontroller controls all the functions such as interfacing/initialising the LCD to display all the desired characters on the screen,

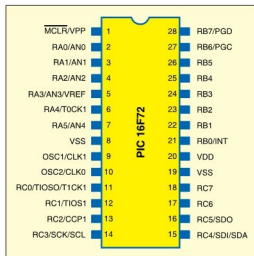


Fig. 1: Pin configuration of PIC16F72 microcontroller in PDIP package

such as banks, casinos, shops, departmental stores and airports or military installations.

Here is a remote-controlled CCTV switcher circuit to monitor six cameras on a single or dual monitor/TV. The salient features of this switcher are:

1. Cameras: Six cameras with one-way audio (audio incoming only)
2. Operation: Fully remote-controlled (infrared commander)
3. Front display: LCD to display the camera selected and all other functions/parameters
4. Camera selection: Manual (you can select any camera of your choice by pressing number keys or camera up/down keys on the remote) and



PARTS LIST

Semiconductors:

IC1	- PIC16F72 microcontroller
IC2-IC4	- CD4066 quad bilateral CMOS switch
IC5	- UTC820 power amplifier
T1-T6	- BC547
IC6	- 78S12 voltage regulator
IC7	- 7805 voltage regulator
X ₁	- 4MHz crystal
D1-D4	- 1N4007 diode
LED1-LED6	- LED, 5mm red
LED7	- LED, 5mm green
IRx	- TSOP1738 IR receiver

Resistors (all 1/4-watt, $\pm 5\%$ carbon unless stated otherwise):

R1	- 10-kilo-ohm
R2, R13	- 2.2-kilo-ohm
R3-R8	- 470-ohm
R9, R10	- 75-ohm
R15	- 47-ohm, 1-watt
R11, R16	- 47-ohm
R12	- 1-ohm
R14	- 1-kilo-ohm
R17-R22	- 1-kilo-ohm
VR1	- 10-kilo-ohm
VR2	- 10-kilo-ohm, log, potentiometer

Capacitors:

C1	- 1000µF, 35V electrolytic
C2	- 220µF, 40V electrolytic
C3, C10, C13, C16	- 100µF, 25V electrolytic
C4, C5	- 33pF ceramic
C6, C8, C14, C18	- 0.1µF mylar
C7, C9	- 47µF, 16V electrolytic
C11	- 680pF ceramic
C12	- 22µF, 25V electrolytic
C15	- 220µF, 25V electrolytic
C17	- 10µF, 25V electrolytic

Miscellaneous:

X1	- 0-15V, 2A secondary transformer
S1	- DPDT rocker switch
SK1-SK6	- 5-pin DIN socket
SK7, SK8	- RCA socket
LCD	- 16x2 LCD with backlight
Remote	- No. 6710V00079B (type LG TV)
L5	- 8-ohm, 6-cm speaker
CON1	- 7-pin bergstrip connector (male/female)
CON2-CON7	- 2-pin SIP connector (male/female)
CON8	- 12-pin bergstrip connector (male/female)